

Going beyond rigidity in tensor categories

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Overview

- ① Monoidal categories and dualities
- ② Module categories
- ③ Results

Rigidity

Definition

- 1 An object c in a monoidal category $\mathcal{C}$ admits a *rigid right dual* $c^\vee$, if there exist morphisms

$$\mathrm{ev}^r : c^\vee \otimes c \rightarrow \mathbf{1}, \quad \mathrm{coev}^r : \mathbf{1} \rightarrow c \otimes c^\vee,$$

satisfying zig-zag relations.

- 2 *Rigid left duals* are defined similarly.
- 3 An object $c \in \mathcal{C}$ is called *rigid* if it admits both a left and a right dual.

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Examples

- 1 Category of finite-dimensional vector spaces.
- 2 Category of finite-dimensional modules over a finite-dimensional Hopf algebra.

Consequences of rigidity

- 1 In abelian rigid monoidal categories monoidal products are **exact**.
- 2 Projective objects form a monoidal ideal.
- 3 Taking duals is opmonoidal: $(c \otimes d)^\vee \cong d^\vee \otimes c^\vee$

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- 1 For A a finite-dimensional associative algebra and $B \in A\text{-Bimod}$. Then $B \otimes_A -$ is generally only **right exact**.
- 2 V a vertex operator algebra and $M \in V\text{-Mod}$ (to which HLZ theory applies). Then $M \otimes_V -$ is generally only **right exact**.

An extended example

A an algebra, $B, \tilde{B}$ bimodules.

- The tensor product $B \otimes_A \tilde{B}$ is constructed by the coequaliser

$$B \otimes_{\mathbb{K}} A \otimes_{\mathbb{K}} \tilde{B} \rightrightarrows B \otimes \tilde{B} \rightarrow B \otimes_A \tilde{B} \rightarrow 0$$

and is hence **right exact**. Monoidal unit is A .

- If B is an A -bimodule, then the linear dual $B^* = \text{Hom}_{\mathbb{K}}(B, \mathbb{K})$ is also an A -bimodule with action characterised by

$$(a_2 \cdot \beta \cdot a_1)(b) = \beta(a_1 \cdot b \cdot a_2), \quad \beta \in B^*, \quad b \in B, \quad a_i \in A.$$

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In general, $A^* \not\cong A$. Smallest example:

$A = \mathbb{K}[x, y] / \langle x^2, y^2, xy \rangle$. Admits a unique simple module S .

$$0 \longrightarrow S \oplus S \longrightarrow A \longrightarrow S \longrightarrow 0.$$

Example continued

A an algebra, $B, \widetilde{B}$ bimodules.

- A^* is a coalgebra and B is an A^* -bicomodule with coactions

$$\begin{aligned}\rho_r : B &\rightarrow B \otimes_{\mathbb{K}} A^*, & \rho_l : B &\rightarrow A^* \otimes_{\mathbb{K}} B. \\ b &\mapsto b.(-) & b &\mapsto (-).b\end{aligned}$$

- The equaliser

$$0 \rightarrow B \otimes^A \widetilde{B} \rightarrow B \otimes_{\mathbb{K}} \widetilde{B} \rightrightarrows B \otimes_{\mathbb{K}} A^* \otimes_{\mathbb{K}} \widetilde{B}$$

defines a **left exact** monoidal product (the cotensor product) with monoidal unit A^* .

- We have $\mathrm{Hom}_A(B \otimes_A \widetilde{B}, A^*) = \mathrm{Hom}_A(B, \widetilde{B}^*)$.

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$$A \otimes_A A \cong A, \quad A \otimes_A A \cong S^{4\oplus}.$$

Grothendieck-Verdier categories / $*$ -autonomous categories

Definition

- ① Let $(\mathcal{C}, \otimes)$ be monoidal. $K \in \mathcal{C}$ is called *dualising* if, for every $Y \in \mathcal{C}$
- $$X \mapsto \mathrm{Hom}(X \otimes Y, K)$$
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is representable. That is, there exists some $GY \in \mathcal{C}$ such that

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and the so defined contravariant functor $G : \mathcal{C} \rightarrow \mathcal{C}$ is an antiequivalence.

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- ② A **Grothendieck-Verdier** (GV) category is a monoidal category $(\mathcal{C}, \otimes)$ together with a choice of dualising object $K \in \mathcal{C}$.

More examples

Proposition [Boyarchenko, Drinfeld '13]

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Theorem [Allen-Lentner-Schweigert-SW, '21]

Let V be a vertex operator algebra and $\mathcal{C}$ a category of V -modules (plus mild technical assumptions).

Then the graded dual V' is a dualising object and $(\mathcal{C}, V')$ is GV (even ribbon GV).

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Theorem [Allen '23]

Let H be a Hopf algebroid with finite-dimensional base algebra A and an invertible antipode S . Then the category of finite-dimensional H -modules is a GV category. A dualising object is given by the vector space dual of the base algebra A .

Important consequences

A GV category $(\mathcal{C}, K)$ admits a second monoidal product:

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Proposition

① – $\otimes Y$ admits a right adjoint (internal Hom):

$$\mathrm{Hom}(X \otimes Y, Z) \cong \mathrm{Hom}(X, \underline{\mathrm{Hom}}(Y, Z)), \text{ with } \underline{\mathrm{Hom}}(Y, Z) \cong Z \boxtimes GY.$$

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- ① $- \otimes Y$ admits a right adjoint (internal Hom):

$$\mathrm{Hom}(X \otimes Y, Z) \cong \mathrm{Hom}(X, \underline{\mathrm{Hom}}(Y, Z)), \text{ with } \underline{\mathrm{Hom}}(Y, Z) \cong Z \boxtimes GY.$$

- ② $Y \boxtimes -$ admits a left adjoint (internal coHom)

$$\mathrm{Hom}(X, Y \boxtimes Z) \cong \mathrm{Hom}(\underline{\mathrm{coHom}}(Y, X), Z), \text{ with } \underline{\mathrm{coHom}}(Y, X) \cong GY \otimes X.$$

Module categories

Motivation

- The construction of two-dimensional conformally invariant conformal field theories (CFT) requires module categories.
- Frobenius algebras constructed from module categories describe (boundary) field objects in CFT.

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- Frobenius algebras constructed from module categories describe (boundary) field objects in CFT.
- Separate mathematical question: GV categories are said to admit distributors:

$$\delta^r : X \otimes (Y \boxtimes Z) \rightarrow (X \otimes Y) \boxtimes Z, \quad \delta^l : (X \boxtimes Y) \otimes Z \rightarrow X \boxtimes (Y \otimes Z)$$

These are associators for mixed tensor products and are generally **not** isomorphisms.

This is clarified in [Fuchs, Schaumann, Schweigert, '23&'24].

Module categories

All results from now on are from [Fuchs, Schaumann, Schweigert, SW '24].

Definition

A *left GV-module category* over a GV category $(\mathcal{C}, \otimes, K)$ is a left module category $(\mathcal{M}, \triangleright)$ such that the functors

$$- \triangleright m : \mathcal{C} \rightarrow \mathcal{M}, \text{ and } c \triangleright - : \mathcal{M} \rightarrow \mathcal{M}$$

both admit right adjoints.

Denote the respective right adjoints by $\underline{\text{Hom}}(m, -)$ and $R_c(-)$.

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Proposition

Let $(\mathcal{M}, \triangleright)$ be a left GV-module category over $(\mathcal{C}, \otimes, K)$, then the functor

$$\blacktriangleright : \mathcal{C} \times \mathcal{M} \rightarrow \mathcal{M}, \quad c \blacktriangleright m = R_{Gc}(m)$$

admits left adjoints when fixing either variable and defines a left module structure over $(\mathcal{C}, \otimes)$.

Distinguished subcategories

The functors $\underline{\mathrm{Hom}}(m, -)$ and $\underline{\mathrm{coHom}}(m, -)$ (left adjoint to $- \blacktriangleright m$) are naturally weak module functors.

Definition

- $m \in \mathcal{M}$ is called $\otimes$ -admissible if
 - 1 $\underline{\mathrm{Hom}}(m, -)$ is a strong module functor over $(\mathcal{C}, \otimes)$.
 - 2 $\underline{\mathrm{Hom}}(m, -)$ admits a right adjoint. $\widehat{\mathcal{M}}^{\otimes}$ is the full subcategory of $\otimes$ -admissible objects.
- $\otimes$ -admissibility is defined analogously using $\underline{\mathrm{coHom}}(m, -)$
- Considering $\mathcal{C}$ as a module category over itself yields $\widehat{\mathcal{C}}^{\otimes}, \widehat{\mathcal{C}}^{\boxtimes}$.

Proposition

- 1 The subcategories $\widehat{\mathcal{C}}^{\otimes}, \widehat{\mathcal{C}}^{\boxtimes}$ are unital monoidal subcategories in $(\mathcal{C}, \otimes)$ and $(\mathcal{C}, \boxtimes)$.
- 2 $\widehat{\mathcal{M}}^{\otimes}$ and $\widehat{\mathcal{M}}^{\boxtimes}$ are module categories over $\widehat{\mathcal{C}}^{\otimes}, \widehat{\mathcal{C}}^{\boxtimes}$ by restriction.

Algebras and coalgebras

For any $m \in \mathcal{M}$,

- Internal End $A_m = \underline{\text{Hom}}(m, m)$ is an associative unital algebra in $(\mathcal{C}, \otimes)$.
- Internal coEnd $C_m = \underline{\text{coHom}}(m, m)$ is a coassociative counital coalgebra in $(\mathcal{C}, \otimes)$.
- Natural module functor

$$\begin{aligned}\underline{\text{Hom}}(m, -) : \mathcal{M} &\rightarrow \text{Mod-}A_m \\ n &\mapsto \underline{\text{Hom}}(m, n)\end{aligned}$$

Same for $\underline{\text{coHom}}(m, -)$ and comodules.

Proposition

If $\mathcal{C}, \mathcal{M}$ are finite abelian, then

- 1 For $m \in \mathcal{M}$, the functor $\underline{\text{Hom}}(m, -)$ is a module equivalence if and only if $m \in \widehat{\mathcal{M}}^\otimes$ and m is a $\mathcal{C}$ generator of $\mathcal{M}$.
- 2 Similar for $\underline{\text{coHom}}(m, -)$.

Frobenius algebras

Proposition

Let $(A, \mu, \eta) \in (\mathcal{C}, \otimes)$

$$\mu : A \otimes A \rightarrow A, \quad \eta : \mathbf{1} \rightarrow A,$$

be a unital associative algebra. Then the following are in bijection.

- ① Coalgebra structures $\Delta : A \rightarrow A \otimes A$, $\epsilon : A \rightarrow K$ such that

$$(\mu \otimes \text{id}_A) \circ \delta_{A,A,A}^l \circ (\text{id}_A \otimes \Delta) = \Delta \circ \mu = (\text{id}_A \otimes \mu) \circ \delta_{A,A,A}^r \circ (\Delta \otimes \text{id}_A).$$

- ② Morphisms $\kappa : A \otimes A \rightarrow K$ such that

$$\kappa \circ (\mu \otimes \text{id}_A) = \kappa \circ (\text{id}_A \otimes \mu) \circ \alpha_{A,A,A}.$$

- ③ Frobenius forms $\lambda : A \rightarrow K$ (related to isomorphisms $A \rightarrow GA$).

Serre functors and Frobenius algebras

Proposition

- ① There exists an equivalence $S : \widehat{\mathcal{M}}^{\otimes} \rightarrow \widehat{\mathcal{M}}^{\boxtimes}$ called the *relative Serre functor* such that

$$\underline{\mathrm{Hom}}(m, -) \cong \underline{\mathrm{coHom}}(Sm, -)$$

as module functors.

- ② The relative Serre functor of $\mathcal{C}$ is canonically a monoidal equivalence $S_{\mathcal{C}} : \widehat{\mathcal{C}}^{\otimes} \rightarrow \widehat{\mathcal{C}}^{\boxtimes}$.
- ③ The relative Serre functor $S_{\mathcal{M}}$ of $\mathcal{M}$ is a twisted module functor

$$S_{\mathcal{M}}(c \triangleright m) \cong S_{\mathcal{C}}(c) \blacktriangleright S_{\mathcal{M}}(m), \quad c \in \widehat{\mathcal{C}}^{\otimes}, m \in \widehat{\mathcal{M}}^{\otimes}.$$

- ④ Let $m \in \widehat{\mathcal{M}}^{\otimes}$. For every isomorphism $p : m \rightarrow Sm$ there is a natural Frobenius algebra structure on $\underline{\mathrm{Hom}}(m, m)$.

Summary

- Grothendieck-Verdier duality is a natural duality structure for many algebraic phenomena.
- There is a natural theory of module categories over Grothendieck-Verdier categories that provides many of the features needed for conformal field theory.
- Next: Try and construct conformal field theories from Grothendieck-Verdier categories and their modules.

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Thank you for coming!